

Ram Ji Collection - Tech Stack & Cloud Report

Project Tech Stack, Database, Cloud & Architecture Summary

Generated Date: Wednesday, 29 July 2026

1. Programming Languages & Web Technologies (Konsi Language Use Ki Hai?)

- **TypeScript:** Entire Backend (Express API) + Frontend (React Web, Admin) + Shared Packages me primary language ke roop me use ki gayi hai.
Kyu Use Kiya: Strict type-checking, code autocomplete, compile-time bugs capture karne aur scalable enterprise codebase ke liye.
- **JavaScript (ES6+):** Runtime scripting, async execution aur node ecosystem ke liye.
Kyu Use Kiya: Node.js backend environment aur browser environment me code execute karne ke liye.
- **SQL & Prisma Schema Definition:** Database tables structure, models, enums aur relations define karne ke liye.
Kyu Use Kiya: Structured data models (Users, Products, Orders, Cart, Reviews) build karne ke liye.
- **HTML5 & CSS3 / Tailwind CSS:** Web page layout structure aur modern UI styling.
Kyu Use Kiya: Responsive, fast-loading, visually attractive e-commerce interface design karne ke liye.

2. Database Engine & ORM (Konsa Database Use Kiya Hai?)

- **PostgreSQL:** Primary Relational Database Management System (RDBMS).
Kyu Use Kiya: E-commerce platform me complex relational data (Users, Cart, Orders, Payments, Inventory) ko secure, fast, aur ACID-compliant tarike se manage karne ke liye.
- **Prisma ORM:** Next-generation Node.js & TypeScript ORM.
Kyu Use Kiya: Direct raw SQL ki bajaye type-safe database queries, schema migrations aur automatic relationship handling ke liye.

3. Cloud Infrastructure & External Cloud APIs (Konsa Cloud Use Kr Rhe Hain?)

- **Neon Postgres Cloud (Database Cloud):** Serverless PostgreSQL Database hosted on AWS Cloud infrastructure.
Kyu Use Kiya: Cloud-hosted database flexibility, auto-scaling, high availability, aur secure SSL database connection ke liye.
- **Cloudflare R2 (Storage Cloud):** S3-compatible Object Storage Service by Cloudflare.
Kyu Use Kiya: Product high-resolution images, banners, apparel thumbnails store karne ke liye without any egress bandwidth charges.
- **Resend API (Email Cloud):** Cloud-based transactional email dispatch service.
Kyu Use Kiya: Order confirmation emails, customer notifications, and invoices send karne ke liye.
- **Meta WhatsApp Cloud API (Messaging Cloud):** Official Cloud API provided by Meta/WhatsApp.
Kyu Use Kiya: Customers ko directly WhatsApp par live order tracking and order updates bhejne ke liye.
- **Razorpay Cloud API (Payments):** Cloud Payment Gateway Integration.
Kyu Use Kiya: UPI, Credit/Debit Cards, Netbanking, aur Wallets ke zariye secure payments accept karne ke liye.

4. Frontend & Admin App Architecture

- **React.js (v18) + Vite:** Modern Single Page Application (SPA) framework for Customer Storefront (apps/web) & Admin Panel (apps/admin).
Kyu Use Kiya: Fast client-side rendering, instant page transitions, aur blazing-fast build speed ke liye.
- **Zustand State Management:** Lightweight client state management.
Kyu Use Kiya: Cart items, user auth tokens, theme states easily manage karne ke liye.
- **Framer Motion & Lucide Icons:** UI animations & vector icons library.
Kyu Use Kiya: Premium visual design aesthetics, dynamic animations aur icons provide karne ke liye.

5. Project Workspace Architecture

- **NPM / PNPM Workspaces (Monorepo):** Multi-package workspace structure (apps/web, apps/admin, apps/api, packages/types, packages/ui).
Kyu Use Kiya: Frontend aur Backend ke bich shared TypeScript interfaces aur UI components bina code duplication ke reuse karne ke liye.